

Manipulate formulas involving several variables using digital tools, and describe the effect of systematic variation in the values of the variables

Learning intention

- manipulate formulas involving several variables using digital tools, and describe the effect of systematic variation in the values of the variables.

Teach and model

- experimenting with different sets of tables of values from formulas.
- using spreadsheets and the formula function to recognise the effect of changing parameters on the entries in cells.
- understanding that the relationship between variables in an artificial intelligence model assists with explainability and interpretability.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.